Comparing Fitness Outcomes of Sexually- and Asexually-Produced Offspring in *Daphnia pulicaria*

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A_m x A_f	A_m x B_f	A_m x C_f
B_m x A_f	B_m x B_f	B_m x C_f
C_m x A_f	C_m x B_f	C_m x C_f

Figure 1: Organizational diagram of proposed *D. pulicaria* crosses. Reproductive controls (inbred groups) are marked in red.
Line A: 3L2-1 Line B: Fish Lake Line C: RLSD 26

Purpose

Calculate inbreeding depression for clones and compare fitness between inbred and outbred groups by statistical analysis of both success and physiological attributes.

Major Theories

1. Asexual and sexual reproduction
2. Genetic crossing
3. Fitness
4. DNA recombination, cloning, and gene conversion
5. Retrotransposons, transposons, and transposable elements (some known in *Daphnia*)
6. Measures of fitness
    a. Juvenile survival: proportion of offspring who survive until reproductive age
    b. Lifetime reproductive success: number of offspring who survive until reproductive age (juvenile survival) per mother; likely necessary to use relative (to ephippia produced by crossed *Daphnia*) LRS
    c. Inbreeding depression: direct comparison of fitness values for inbred and outbred individuals of the same species

$$\delta = \frac{W_{outbred} - W_{inbred}}{W_{outbred}}$$

    d. Heterozygosity-fitness correlation: the impact that heterozygosity (at >1 loci) has on the fitness of an individual; correlated with reproductive success or survival

Experimental setup

Cross Group (Experimental)

1. Gathering materials for crosses:
    a. Setup 4 beakers for each line (12 total). Mark each stock culture with its respective clonal line name.

b. Gather *Daphnia* from each clonal main line and isolate 6-8 individuals per beaker. Ensure that >2 females and >1 male are present in each beaker.
c. Feed stock cultures with algae solution 4-5 times per week.
d. Allow the cultures to grow until roughly 50 *Daphnia* are present in each beaker.

2. <u>Isolating *Daphnia* for crosses:</u>

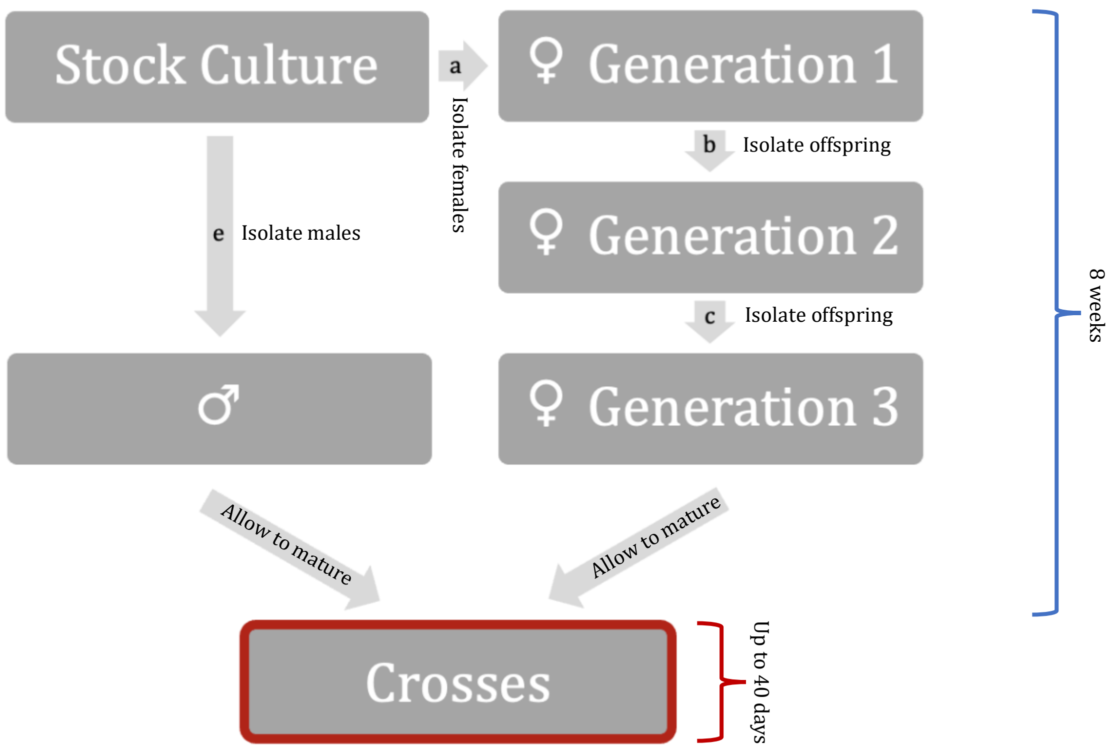

Figure 2: Diagram of estimated time for collection, isolation, and maturation steps of *D. pulicaria* crosses.

a. Immediately being isolating females (per line, 3 beakers of 2 females each).
b. Allow the females to grow and produce clones, then gather the virgin female clones in the same manner as in the previous step.
c. Allow these 2nd generation virgin females to grow and produce clones, then gather the virgin female offspring (which will be used in the crosses).
d. Allow these 3rd generation virgin females to mature before setting up the crosses.
e. Gather mature males from each clonal line, isolating them into new cultures with 150 mL of water and marking each new culture with its respective clonal line name.
f. Criteria for beginning crosses:
    i. 15 mature males per clonal line
    ii. 50 3rd-generation virgin females per clonal line

3. <u>Crosses:</u>
    a. Overview:
        i. 9 crosses, including three reproductive control (inbred) groups
        ii. 5 cultures per cross
            1. 10 virgin females, 3 mature males per cross
            2. n=15 males per cross; n=50 females per cross
    b. Materials:
        i. 45+ small beakers
        ii. 15 mature males from each clonal line
        iii. 50 virgin females from each clonal line
    c. Setting cross cultures:
        i. Into each new beaker for all 9 crosses, add 3 mature males and 10 virgin females.
        ii. Controlled variables:
            1. Photoperiod (standardized for all sets, all crosses)

    d. Gathering data:
        i. In the weeks following cross setup, check crosses for ephippia and, if ephippia are present, check ephippia for eggs. Findings should be noted in a table in the Lab Notebook.
            1. Ephippia (that contain eggs) are the only evidence of cross success. Separate ephippia into marked centrifuge tubes.
        ii. In the process of collecting ephippia, remove mature males and virgin females to new beakers after each offspring cycle. Use the same generation of males and females for the entirety of the experiment.
            1. Allows for experiment continuation and more data to collect.
            2. Continue checks through clutch 6.
4. Measurements and standards:
    a. Stock, sex-isolated, and crossed cultures require 150 mL of water each, with water being added weekly to replace any that evaporates.
    b. Photoperiod for all cultures: standard 12-hour photoperiod during gathering period; switch to 10-hour daylight period for crosses.

Clone Group (Control)

1. Allow 4 stock cultures per inbred group (12 cultures total) to grow for the entire setup and collection periods of the cross experiment.
2. To minimize contamination risk, reset stock cultures at least once per week, keeping 6-8 *Daphnia* per beaker during each reset.
3. Whenever during the experiment is most convenient, set new beakers of cloned *Daphnia* females: 5 beakers of 10 females each, per clonal line.

Data Analysis: Cross Group

1. Gather all ephippia from centrifuge tubes and remove the eggs.
2. Attempt to hatch offspring from the eggs
    a. MORE INFORMATION NEEDED
3. Place hatched offspring in new beakers, marked with their respective crosses.
4. Count and make note of the number of hatched Daphnia for each cross.
5. Allow these *Daphnia* to grow in a culture until they die. Replace the beakers weekly, but attempt to preserve all offspring.
6. Check and count Daphnia daily, and track deaths.
7. Determine juvenile survival from the proportion of living Daphnia in each cross between 5 and 10 days.
8. Determine relative LRS from the number of living Daphnia in each cross between 5 and 10 days.
9. Determine fitness values (days lived) for crossed offspring and make a note to prepare for inbreeding depression calculation.

Data Analysis: Clone Group

1. With the new beakers of isolated clones, follow steps 5-9 of previous subsection (*Data Analysis: Clone Group*).
2. Calculate inbreeding depression (see **Major Theories**, section 6c).

Complications/Questions

1. Are 5 replicates necessary for each cross to ensure statistical significance (see 3a)?
2. Should I set a specific amount of time (# of days) for the juvenile survival and LRS collection periods? What number of days between 5-10?
3. What are the most important statistical comparisons to make when analyzing data? Are multiple methods helpful/necessary?

Continued Research Ideas

1. How do methyl farnesoate and N-methyl-D-aspartate (NMDA) influence the sex of daphnia produced under mating-season conditions?

a. What modulators/ligands of the *Daphnia* NMDA receptors can influence reproductive methodology and outcomes when methyl farnesoate is present among reproducing individuals (Toyota et al. 2015)?
b. If offspring are produced, are males and females (respectively) of similar fitness to those not produced by parents exposed to methyl farnesoate (control group)?
    i. For experimentation in which males are specifically required, does the use of methyl farnesoate (MF) and NMDA affect the natural fitness of *Daphnia* offspring? Can researchers use MF and NMDA in the performance of other *Daphnia* experiments without affecting natural selection?
c. Theoretical experimental setup:

Conditions	Group 1	Group 2	Group 3
10-hour daylight period	☒	☒	☒
10° C air temperature	☒	☒	☒
Treatment with methyl farnesoate	☐	☒	☒
Treatment with NMDA	☐	☐	☒

2. Does the use of MF and NMDA in genetic crosses of *Daphnia pulicaria* affect the magnitude of inbreeding depression (ID) among crossed offspring?
    a. Results from the current genetic cross experiment will be crucial for answering this question.
    b. Results from the first Continued Research Idea will also be helpful; the resulting paper can be composed of 3 parts:
        i. *D. pulicaria* genetic cross experiment (including ID measurement)
        ii. *D. pulicaria* MF and NMDA experiment
        iii. D. *pulicaria* genetic cross experiment with MF and NMDA (including ID measurement)
3. Can MF and NMDA be used to produce male offspring in species of Daphnia that do noy typically produce males?